PLANRS ACADEMY

Junior Engineer Series: Grade 3 Science

Friction

Brake Pads Investigation

Session 1: The Stop Secret

The Great Escape!

Imagine you are riding your cycle as fast as you can down the colony road. Suddenly, a cow steps right in front of you! What do you do? You squeeze your brakes hard, and — Screeeech! You stop just in time. How did your cycle know how to stop? Today, we investigate the invisible force that saved the day!

Our Mission Today

As Junior Engineers, we are going to crack the code on friction. Our goals are:

- ⚙️ **Define Friction:** Understand the "invisible grip" between surfaces.
- ⚙️ **Investigate Brake Pads:** See how cycles use friction to stay safe.
- ⚙️ **Rough vs. Smooth:** Discover why the surface of an object matters.
- ⚙️ **Helpful Friction:** Find everyday examples of friction in our Indian homes.

🧼 **WARM-UP ACTIVITY: THE HAND RUB**

Press your palms together tightly and rub them back and forth quickly. What do you feel? Do you feel resistance? Do they get warm? That "grip" and heat is **Friction!**

What is Friction?

Friction is a force that acts like a "speed bump" for moving things. It happens whenever two surfaces rub against each other.

Definition: Friction is a force that opposes motion. It tries to slow down or stop objects that are sliding or rolling past each other.

The Invisible "Grip"

Friction depends on how "rough" or "smooth" a surface is. Rough surfaces have more bumps that "hook" into each other, creating a stronger grip.

🛣️ **COLONY ROAD TEST:**

Think about walking on a smooth, soapy bathroom floor versus a rough cement road. Where is it easier to slip? On the soapy floor, there is low friction. On the rough road, there is high friction that helps your Chappals grip the ground!

Investigation: The Brake Pad Secret

Look at your bicycle. When you pull the brake lever, something happens near the wheels. Let's look closer at the **Brake Pads**.

What are they?

Brake pads are small blocks of tough rubber. They are positioned right next to the metal rim of your cycle wheel.

How they work:

1. You squeeze the lever on the handlebar.
2. A cable pulls the brake pads inward.
3. The rubber pads press hard against the spinning metal wheel rim.
4. The **Friction** between the rubber and the metal slows the wheel down until it stops!

[Visual: Illustration of a brake pad hugging a cycle wheel rim]

Why Rubber? (Rough vs. Smooth)

Why do cycle companies use rubber for brake pads instead of something smooth like silk or plastic?

Rubber is a high-friction material. It is "grippy." If we used smooth plastic, the pads would just slide over the wheel, and you wouldn't be able to stop in time!

> 🎯 **The Carrom Board Mystery:**
>
> Think about playing Carrom. When the striker doesn't slide well, what do you do? You sprinkle powder on the board! The powder fills the tiny bumps on the board, making it smooth and reducing friction so the striker zooms across.

💡 **JUNIOR ENGINEER QUESTION:**

Would you want friction on a Carrom board? No! But would you want friction on your bicycle brakes? **YES!**

Friction: Our Everyday Hero

Without friction, our world would be a very slippery place! Here is where we find it in India:

🏃 **1. Walking and Running:**
Your Bata Chappals or school shoes have patterns on the bottom. These patterns create friction with the ground so you can push off and run without sliding.

✏️ **2. Writing with a Pencil:**
Friction between the lead of your pencil and the paper rubs off bits of graphite, leaving behind your beautiful handwriting.

🧯 **3. Lighting a Matchstick:**
When you rub a matchstick against the rough side of the box, the friction creates enough heat to start a fire for the Puja or the stove.

Helpful or Harmful?

Sometimes we want friction, and sometimes we don't. As an engineer, you have to decide!

Scenario	Friction Type	Why?
Bicycle Brakes	**Helpful**	To stop safely.
Carrom Board	**Harmful**	Slows the striker.
Walking on Tiles	**Helpful**	Prevents slipping.
Rusty Door Hinges	**Harmful**	Makes a loud noise!

Try it Yourself: The Slider Test

You can test friction at home with this simple investigation.

Materials:

📦 A small toy car or a plastic bottle cap.
📦 A smooth tray or a wooden board.
📦 Different surfaces: A cotton cloth, a piece of sandpaper, and some oil.

The Test:

1. Tilt the board slightly to make a ramp.
2. Let the object slide down the smooth board. How fast was it?
3. Now, cover the board with the cloth and try again. Did it slow down?
4. Finally, try the sandpaper. What happened?

> 📊 **Result: The sandpaper has the most friction, so the object moves the slowest!**

Engineer's Safety Note

Investigating machines is fun, but safety is the most important part of any science experiment.

🛡️ **SAFETY CHECKLIST:**

• **Moving Parts:** Never put your fingers near the cycle wheel while someone else is spinning it. You could get pinched!

• **Hot Brakes:** After a long ride, brake pads can get very warm because of friction. Don't touch them right away!

• **Adult Supervision:** Always have an adult help you when you are examining your bicycle's mechanical parts.

Summary: The Stop Secret Revealed!

You have successfully investigated friction! Here is what we learned:

✨ **Friction** is the force that tries to stop things from moving.
✨ **Brake Pads** use rubber to create high friction against the wheel rim.
✨ **Rough surfaces** have more friction than smooth ones.
✨ Friction is **helpful** for stopping, walking, and writing, but **harmful** when we want things to slide easily.

Great Job, Junior Engineer!

Now you know the secret force that keeps us safe on the road. Keep investigating!